



miniMOT V2 USER MANUAL



©Infleqtion 2025

1315 Century Drive
Suite 150
Louisville, Colorado
303-440-1284

sales@infleqtion.com
www.infleqtion.com



Table of Contents

1.	General Information	3
1.1	User Facilities Requirements	3
1.2	Package Contents.....	3
2.	Safety	3
2.1	Definitions, Labels and Symbols	3
2.2	Warnings	4
2.3	Physical Safety.....	4
2.4	Electrical Safety	4
2.5	Laser Safety	4
3.	Specifications	5
4.	Regular Product Maintenance	5
4.1	Handling	5
4.2	Cleaning.....	6
5.	Set-up	6
5.1	Mounting.....	6
5.2	System Start	8
5.3	Dispenser Validation	8
5.4	Coil Validation and Final System Checks.....	8
6.	Operation	9
6.1	Normal Operation	9
6.2	Turning the miniMOT V2 off	9
6.3	Creating a MOT	10
7.	Troubleshooting.....	10
8.	Warranty	11
9.	Technical Drawings and Dimensions	12
10.	References and Further Reading	13
11.	Appendix: Principles of Operation.....	13
11.1.1	The Ultrahigh Vacuum Cell and Ion Pump	14
11.1.2	The Alkali Metal Dispenser	15
11.1.3	Accessories.....	15



1. General Information

1.1 User Facilities Requirements

Ensure your lab or experimental setup has ample room to work around the cell for installation and laser safety. Always follow your institutional guidelines concerning a safe work environment.

1.2 Package Contents

Upon receiving the miniMOT V2, inspect the packaging for damage. If the packaging shows signs of damage, excessive shock or if the shock watch is red, notify the shipping company and then contact Infleqtion.

When unpacking the miniMOT V2, be particularly careful to avoid dropping it or knocking it against anything. Carefully remove the miniMOT V2 and all the accessories and inspect for any damage. Please contact Infleqtion if there are signs of damage to the product. Keep all original packing materials.

Upon receipt of your miniMOT V2 product we advise that the package is opened, and the unit is powered ON.

The shipment should contain the following items:

- A miniMOT V2 (with integrated vacuum cell)
- An AC/DC power adapter and power cord
- Anti-Helmholtz pair of magnetic field coils

Note: the magnetic field coils are mounted to the miniMOT V2 prior to shipping

If any of these items are missing, please contact Infleqtion to obtain replacements.

2. Safety

2.1 Definitions, Labels and Symbols

Symbol	Description
	Caution! – Risk of electric shock
	Caution! – Risk of damage to equipment
	Caution! – Risk of burn due to hot surface



2.2 Warnings

	SHOCK WARNING - To avoid electric shock, NEVER open the enclosure without consulting Inflection first. Opening the enclosure without approval from Inflection voids the warranty.
	NOTICE – Before using the miniMOT V2 for the first time, follow the instructions in Section 4.1 “Unpacking the miniMOT V2” to ensure the system is functioning properly. If the miniMOT V2 is not functioning properly, contact Inflection at sales@inflection.com .
	HOT SURFACE WARNING – If used with Inflection’s miniMOT coils (MOT Coils), avoid physically touching the coils as they can reach 60°C under normal operating conditions.

2.3 Physical Safety

DANGER!	If you are wearing magnetic implants, take care to keep away from the magnets. Always warn persons wearing such implants of the presence of the magnetic field. Persons with any magnetically sensitive implants should consult their medical doctor regarding any potential applications.
DANGER! CAUTION!	The magnetic fields that extend beyond the protective housings of the device(s) can attract other magnetic materials. Keep small ferromagnetic objects like tools, razor blades, or screws away from the devices.

2.4 Electrical Safety

The ion pump always operates at a fixed voltage of 3.6 kV. To avoid electrical shock, it is recommended that users **do not open the enclosure** without consulting Inflection.

Note: Opening the enclosure void the product warranty.

2.5 Laser Safety

Disclaimer	This is not a laser product. This is for reference only. Please refer to the manufacturer manual and safety information for each laser system that you will use with this product. Please follow your institutional guidelines concerning laser safety.
DANGER!	If the equipment is used in a manner not specified by the manufacturer, then the protection provided by the equipment may be impaired.

This product has specular surfaces, and reflections of the laser may occur. Laser safety glasses should be worn when using this product with a laser. Please follow your institutional guidelines concerning laser safety.



3. Specifications

Parameter	Conditions	Min	Typ.	Max	Unit
External Dimensions	$h \times w \times d$	149.1 $\times$ 119.4 $\times$ 298.5			mm
Nominal Cell Height	At cell center	88.9			mm
Weight			3.2	5	kg
Electrical:					
Single phase					
AC Power Supply Input Voltage		100-240			V
AC Power Supply Input Power		20		<60	W
AC Power Supply Input Frequency		50 - 60			Hz
DC Power Supply Output Voltage		11.8	12	12.6	V
DC Power Supply Output Current			<2	5.0	A
Instrument Load Current	Dispenser Off Coil Current = 0.5A HV supply On	0.6			A
Instrument Load Current	Dispenser On = 4A Coil Current = 1A HV Supply On	1.7			A
Ion Pump Current		<200		1000	nA
Vacuum Quality with No Rb/Cs Load		<10			nTorr
Residual Magnetic Field from Ion Pump	At cell center	<0.5			G
*Clear Aperture					
Sides of Cell	$h \times w$	45 x 10		54 x 19	mm ²
End of Cell	$\varnothing$	8		10	mm
Alkali Metal Source	Rubidium			Cesium	
Dispenser Operating Current		0	2.75 - 3.75	4	A
Magnetic Coil Current		-1		1	A
Magnetic Field Gradient (along coil axis)			13 – 14		G/(A-cm)

4. Regular Product Maintenance

4.1 Handling



Take care when handling the miniMOT V2, and the fragile glass cell can be easily chipped or cracked.

4.2 Cleaning

Inflection recommends cleaning the cell in a typical optics manner.

- Appropriate cleaners include IPA, methanol, acetone, or deionized water.

Use a fresh section of wipe each time to avoid glass chips dragging across the coated surface.

5. Set-up



Figure 1: miniMOT V2 overview diagram showing all key product features

5.1 Mounting

The miniMOT V2 can be mounted to any standard optical table. Before using the miniMOT, please mount it securely using the screw threads located on the base of each side. There are screw threads for both imperial and metric screws.

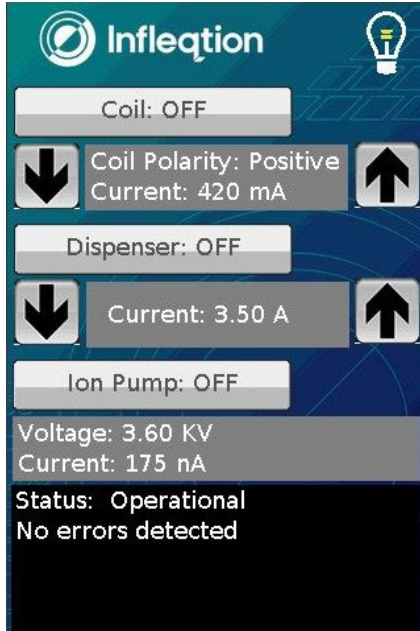


Figure 2. Example of information displayed on the miniMOT V2 touch screen under normal operating conditions.

	Name	Function
1	<i>LCD Display</i>	The LCD display is a touch screen and allows controlling all the miniMOT V2 functionality through the screen. The brightness of the LCD display has 3 levels and can be toggled between them by touching the light bulb in the upper right corner of the LCD.
2	<i>Coil Current Control</i>	Adjusts the magnitude of the coil current between 0 and 1 A. The up and down arrows allow setting the current higher or lower.
3	<i>Coil Current Indicator</i>	Coil “OFF” or “ON” is displayed.
4	<i>Coil On/Off and Polarity Control</i>	Touching the “Coil Polarity: Pos” toggles between negative and positive current flow. When the current in the coil is off, it will be displayed as, “Coil: OFF” After touching on the text, it will toggle to show “Coil: ON”
5	<i>Ion Pump Indicator</i>	Ion Pump indicator “OFF” or “ON” is displayed. Typical current levels are < 200nA.
6	<i>Ion Pump On/Off Control</i>	When the display shows “Ion Pump: ON” the pump is turned on to 3.6kV. Touching the text will toggle the ion pump off and will show the text, “Ion Pump: OFF”
7	<i>Dispenser Current Control</i>	Adjusts the dispenser current between 0 and 4 A. The up and down arrows allow setting the current higher or lower.
8	<i>Dispenser Indicator</i>	Illuminates bright green when current flows through the dispenser.
9	<i>Dispenser On/Off Control</i>	When the dispenser is off, the LCD text shows, “Dispenser: OFF” After touching this text, the screen toggles to display, “Dispenser: ON”



5.2 System Start

To start the miniMOT V2 for the first time, the following steps must be completed in order:

1. Affix the miniMOT V2 securely to a surface before plugging in the power.
2. Plug in the power supply.
3. Locate the power button as shown in Figure 1 and turn on the miniMOT V2.
4. Wait for the main touch screen to display.
5. Locate the ion pump button on the touch screen.
6. Turn on the ion pump and observe the voltage rise to 3.6 kV and the current rise to some non-zero (nA) value.

Within approximately 30 to 60 seconds, you should observe the current value drop to less than 100 nA. This indicates that the ion pump is functioning properly, and the device is under vacuum.

If pump down takes more than 10 minutes, please contact us at sales@inflektion.com while continuing to keep the ion pump operating. In some instances, after a long storage period, the vacuum system could take as long as 48 hours to pump down.

5.3 Dispenser Validation

A final fluorescence test of the miniMOT V2 was completed prior to shipping the unit. It is useful to repeat this test upon receipt of the unit to provide additional validation of the system's vacuum integrity and dispenser connection. This test assumes that you have a laser system that provides laser light resonant with one of the major transitions used for your atom (e.g., 780.2 nm or 852 nm for the D2 transitions of rubidium and cesium respectively).

1. Direct/guide a laser beam of 1-3 mW of power with a beam diameter of 1-3 mm and through the glass cell while scanning the laser frequency over a known, strong atomic/electronic transition.
2. Setup an infrared sensitive camera or infrared viewer to observe fluorescence. A darkened room is advised for optimal viewing of NIR fluorescence.
3. Turn on the dispenser. The touch screen button above the ion pump button activates the dispenser. The arrow buttons below the dispenser button allow you to adjust the dispenser current. The *minimum* current at which you should see fluorescence is 2.75 A and the maximum current at which you should see fluorescence is 3.75 A.
4. Set the dispenser current to the value indicated on the unit's datasheet provided with the unit when shipped and wait 30 minutes before checking for fluorescence. If you do not observe fluorescence, increase the dispenser current by 0.2 A and wait an additional 30 minutes to check for fluorescence.
5. Continue this process until fluorescence is observed, but do not exceed 3.75 A.

The 30-minute intervals allow the cell to be "seasoned" with atomic vapor.

5.4 Coil Validation and Final System Checks

Set miniMOT V2 coil current to 1.0 A and confirm that rear panel display shows 1 Amp actual output. Switch polarity and verify output at -1.0 A. The corresponding orientation of the magnetic field is displayed on Figure 3. A compass can be used to verify the proper switching of



the orientation with polarity.

If you see any error message at the bottom of the display screen, immediately contact Infleqtion at sales@infleqtion.com.

6. Operation

6.1 Normal Operation

NOTE: It is recommended to leave the unit powered and the ion pump on when not in use to maintain appropriate vacuum. If this is the case, proceed to step 4. If not, start with steps 1, 2 and 3.

1. Connect the power cable to the *Power Input* jack on the rear panel of the miniMOT V2.
2. Plug the power adapter into an AC supply per the specification in Table 1. If a different power supply is used, it must match the requirements in Table 1. Improper use will void the warranty.
3. Using the display, set the *Ion Pump On/Off* to the “ON” state. Wait for the *Ion Pump Indicator* to turn bright green.
4. Using the display, set the *Dispenser On/Off* to the “ON” state.
5. Using the display, set the *Dispenser Current* to between 2.5 A to 3.5 A for normal operation. Higher currents may be required to load the cell with rubidium atoms/vapor if the dispenser has been set to OFF for extended periods (see Principles of Operation for more information).
6. To run current through the coils, using the display, set the *Coil On/Off* to the “ON” state. Then set the *Coil On/Off/Polarity* to either “pos” or “neg” direction (depending on laser polarization). Set the coil current recommend starting value of 0.5A, although MOTs can be easily created with currents as low as 0.2 A and as high as 1.0A depending on the magnetic environment in the near vicinity of the product.

Nominal dispenser current for MOT generation: 2.75 - 3.75A

Warning/Error Message	Notes
<i>Dispenser not enabled, ion pump overcurrent.</i>	Normal operation can produce current levels above 200 nA. The dispenser is automatically turned “OFF” to protect the ion pump and the glass cell.
<i>System is not calibrated.</i>	Return to factory for calibration.

6.2 Turning the miniMOT V2 off

1. When not in use, the *Dispenser On/Off* button should put into the OFF state using the touchscreen display.
2. Turn off the coils by moving the *Coil On/Off/Polarity* to the OFF state using the touchscreen display.
3. It is **recommended to leave the ion pump ON at all times**, even over many months. This will maintain the vacuum and there is no wear and tear on the ion pump.



6.3 Creating a MOT

To simplify the production of a 3D MOT, Inflection offers an anti-Helmholtz coil pair (MOT Coils) to create the required magnetic fields, as well as an opto-mechanical (miniMOT V2 Package) for generating a 3D MOT. The MOT Coils are included with the miniMOT V2; the opto-mechanical package is offered as an upgrade from the miniMOT V2.



WARNING!

DO NOT OPEN THE MINIMOT – To avoid shock, NEVER open the enclosure without consulting Inflection. Opening the enclosure without approval from Inflection will void the warranty.

7. Troubleshooting

Problem	Solution
The ion pump Indicator current remains high ($>200\text{nA}$) for more than 24 hours while the dispenser is off	Note the appearance of the glass cell, in particular look for white, blue, purple, or silver films or coatings on the walls. Contact Inflection for assistance at sales@inflection.com . A failed vacuum is likely. This condition is to be avoided when the user operates the product.
The ion pump indicator current is $<200\text{nA}$, and you are unable to observe a MOT	The problem is likely associated with the external optics and photonics used (e.g., laser polarizations, magnetic field orientation, etc.) than with the miniMOT V2.
Nonfunctioning Indicators	Check the power connections and ensure the <i>Ion Pump On/Off</i> is in the “ON” state. If the <i>Ion Pump Indicator</i> remains dark green, unplug the power connector from the miniMOT <i>Power Input</i> and measure the DC voltage across the coaxial connections. It should read 12 VDC. If it does not, check that there is power at the outlet. If power is supplied to the miniMOT V2 but the indicators remain inoperative, or if the power module has failed, contact Inflection for assistance.

Frequently Asked Questions

Can you configure custom hardware for higher MOT atom count?



Yes, the miniMOT can be configured with a custom “L” cell that would lead to a higher atom density. This would be considered a custom product. Please contact Infleqtion for more information.

Can you create an optical tweezer or lattice with the miniMOT?

Yes! The MOT lifetime may be relatively short, on the order of ms.

Can I make a BEC in a miniMOT V2?

No. The pressure is not low enough and since there is only one cell/science chamber, there is no way to separate the “hot” atoms from the dispenser and the colder atoms in the MOT.

Can I make a 3D MOT in a miniMOT V2?

Yes!

Are Rydberg applications possible in the miniMOT V2?

Yes, however there are limitations related to the lifetime and coatings available for wavelengths related to Rydberg work. Please contact Infleqtion for more information.

What power of laser do I need for the miniMOT V2?

The miniMOT V2 requires at least 10 mW of power per beam, and recommends 20 mW per beam.

What temperature are the atoms in a miniMOT V2?

The atoms in a miniMOT V2 are typically in the mK range, typically 100-200 μ K.

8. Warranty

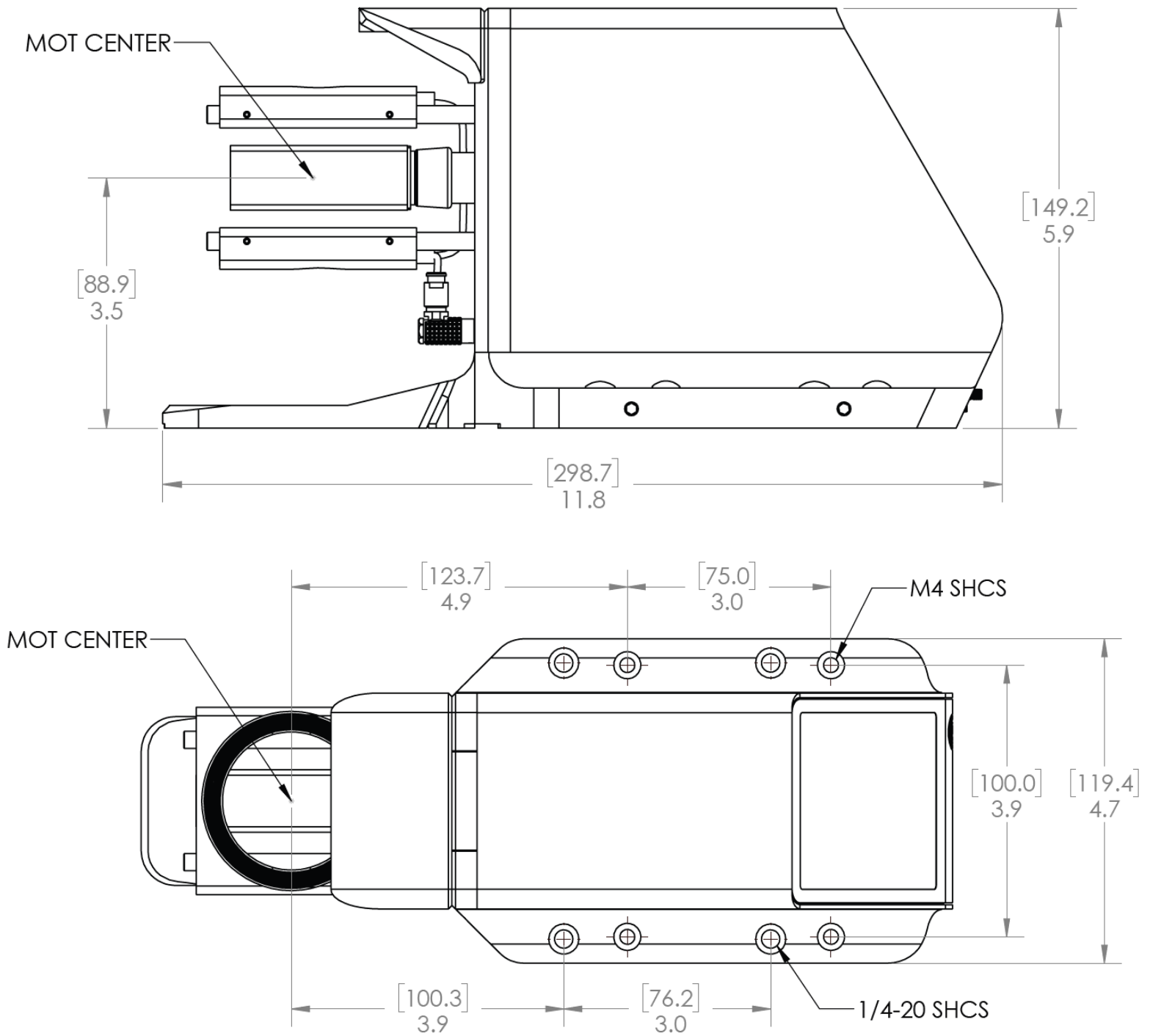
Infleqtion’s Terms and Conditions, including the warranty, can be found at:

<https://infleqtion.com/terms>



9. Technical Drawings and Dimensions

Measurements in mm and [inches]



UNCONTROLLED SALES DRAWING



10. References and Further Reading

- Thurber-Carbone, Brahmin. (2021). Realization of BSU First Magneto-Optical Trap for the Spatial Confinement of Rb Atoms Using Next Generation Fiber Optic Capabilities with MiniMOT. In BSU Honors Program Theses and Projects. Item 497. Available at: https://vc.bridgew.edu/honors_proj/497
- Melody, Thomas. (2024). The Rubidium Magneto-Optical Trap: Designing a Robust System For Measurement & Control. In BSU Honors Program Theses and Projects. Item 671. Available at: https://vc.bridgew.edu/honors_proj/671
- Isichenko, A., Chauhan, N., Bose, D. *et al.* Photonic integrated beam delivery for a rubidium 3D magneto-optical trap. *Nat Commun* **14**, 3080 (2023). <https://doi.org/10.1038/s41467-023-38818-6>
- Xu-Dong Liu, Zong-You Chen, Wei-Chun Huang, Yang-Kao Lee, Yong-Fan Chen, and Pei-Chen Kuan Phys. Rev. A **109**, 013327 – **Published 31 January, 2024** DOI: <https://doi.org/10.1103/PhysRevA.109.013327>

11. Appendix: Principles of Operation

The miniMOT V2 is an integrated vacuum system with optical access for trapping and cooling atoms. The system includes an integrated atomic source for rubidium (Rb) or cesium (Cs) atoms, ion pump and getter pump to maintain vacuum in the chamber, and electronics for driving the vacuum pumps (ion and getter), atomic source, and a pair of magnetic coils (MOT Coils) if ordered. The product is sealed and shipped under vacuum ready to achieve, when properly integrated with the required laser sources, a rubidium or cesium magneto-optical trap (MOT).

The miniMOT V2 consists of:

- Ultrahigh vacuum (UHV) glass cell
- Ion pump
- Passive getter pump
- Alkali metal dispenser (Rb or Cs)
- Electronics to drive the ion pump, the dispenser, and the magnet coils

In conjunction with an external magnetic field gradient and laser beam(s), the miniMOT V2 can be used to produce a three-dimensional (3D) magneto-optical trap from an alkali metal vapor of Rb atoms or Cs atoms. The vapor of atoms is created by running current through the dispenser. To simplify the production of a 3D MOT, Infleqtion offers an anti-Helmholtz coil pair (MOT Coils) to create the required magnetic fields, as well as an opto-mechanical miniMOT V2 Package) for generating a 3D MOT.

Note: Infleqtion does not guarantee MOT creation or product performance when custom magnet coil assemblies are used in conjunction with the miniMOT V2.

The selection of the UHV glass cell is determined at the time of order for a given miniMOT V2 configuration and cannot be changed once the unit is assembled. Several options for UHV glass cells that can be used the miniMOT V2 are highlighted in Table 2.



Table 2. Compatible Glass Cell Options for the miniMOT V2.

UHV Glass Cell	AR Coating Type	miniMOT Product Code	
		Rb Atom Source	Cs Atom Source
2cm Thin-walled cell	Uncoated	CMMR0001	CMMC0001
2cm AR Coated Cells	Uncoated	CMMR0005N	CMMC0005N
	Targeted (H) coating	CMMR0005H	CMMC0005H
	Broadband (K) coating	CMMR0005K	CMMC0005K
High-NA Imaging Cells	Uncoated	CMMR0003N	CMMC0003N
	Targeted (H) coating	CMMR0003H	CMMC0003H
	Broadband (K) coating	CMMR0003K	CMMC0003K

At the heart of the miniMOT V2 is a permanently sealed ultrahigh vacuum (UHV) glass cell. A small, 0.4 liters per second, ion pump maintains the ultrahigh vacuum, while a dispenser supplies rubidium (Rb) or cesium (Cs) atoms to the cell.

11.1.1 The Ultrahigh Vacuum Cell and Ion Pump

The ion pump incorporates a pair of rare-earth magnets. Although these magnets are shielded, there remains a small residual magnetic field. In the MOT region (i.e., the center of the glass cell), this residual field is less than the Earth's field. For most users this will have no impact. The ion pump always operates at a fixed voltage of 3.6 kV. To avoid electrical shock, it is recommended that users **do not open the enclosure** without consulting Infleqtion.

Note: Opening the enclosure will void the product warranty.

When the ion pump is "ON" and the current is greater than 200 nA, then the dispenser cannot be turned "ON". In addition, if the dispenser is set to "ON" and the ion pump current goes above 200 nA, then the dispenser automatically shuts "OFF". Both cases are to protect the ion pump and the cell from experiencing too much dispensed atom vapor.

Note that the current-to-pressure conversion for ion pumps is almost always made with reference to nitrogen pressure. The actual value of this conversion depends upon the gas species being pumped.

When the unit is first switched to the "ON" state, the *Ion Pump Current will be 100s of nA and pump down to 0 nA* in a timeframe of a few seconds to approximately 60 seconds. The actual time will (1) **depend on the length of time the ion pump has been inactive**, (2) the recent rubidium load, and (3) the thermal history of the environment the unit has been used in.

An ion pump can have a difficult time starting when the vacuum pressure is too low because it operates through an electrical discharge, and to create that discharge there must be sufficiently small, yet finite pressure. This condition can arise if the ion pump has been active for a considerable time, but the rubidium dispenser has not.

If the ion pump is deactivated and then turned on again shortly afterwards, the *Ion Pump Indicator* can display green indicating no current flowing through the ion pump because it hasn't started. When the pressure rises sufficiently, either because of background outgassing or dispenser activation, the Message Window on the display will show one of two possible problems



detailed in Table 4 below. The following lists various warning and error messages and notes about the conditions.

11.1.2 The Alkali Metal Dispenser

The dispenser releases rubidium or cesium with the natural abundance of isotopes: 72% ^{85}Rb and 28% ^{87}Rb or 100% ^{133}Cs .

It does so by heating the dispenser element with an electrical current. A safety feature in the miniMOT V2 prevents current from flowing to the dispenser unless the ion pump is “ON”. Dispenser current can be adjusted from 0 A to 4 A.

For most users, a dispenser current in the range of 2.75 to 3.75 A is required to release enough Rb or Cs atoms without overloading the ion pump.

In a new vacuum chamber, the chamber walls act as a pump for the alkali atoms. This means that it can take several hours (>5 hours in some cases) of operation of the dispenser before a significant amount of alkali (Rb, Cs) atoms are visible in the vacuum chamber. We refer to this as “seasoning” of the vacuum. It is normal and should not be cause for concern.

11.1.3 Accessories

Magnetic Field Coils

The miniMOT V2 can be purchased with Infleqtion’s standard magnet Coils (MOT Coils) This assembly, if ordered, comprises of a pair of circular magnetic field coils that are positioned on the top and bottom of the miniMOT V2 glass cell as shown in Figure 2. The coils are wired in series in opposing senses, thereby forming an anti-Helmholtz pair. When driven with a DC current, the coils generate a magnetic quadrupole field whose center coincides with the spatial center of the glass cell. Please see Figure 3 below.

The DC resistance of the two coils in series is approximately 8 Ω . The total inductance of the coils in series in a quadrupole configuration is approximately 5-6 milli Henri (mH).

The coils electrically connect to the miniMOT V2 panel via a 2-pin cable. To flow current through the coils, use *the Coil On/Off/Polarity* controls on the LCD screen. No current flows when the LCD screen shows “Coil OFF”. Current flows when the user touches the “Coil OFF” and then it toggles to show “Coil ON”. The user can then touch/press the up and down arrows to select the current level. The polarity or direction of the current can be changed to be “pos” for positive or “neg” for negative. When *Coil polarity pos*, is shown then a current flows through the top coil first, and then the bottom coil second. When *Coil polarity neg*, is shown then current flows through the coils in the opposite way, i.e., through the bottom coil first and then the top coil. The corresponding orientation of the magnetic field is shown in Figure 3.

The user can set the coil current to any value between 0 A and 1.0 A. Typically, 0.5-0.7 A will produce a MOT. For both current polarities, the coils generate a magnetic quadrupole gradient field of up to 13.5 G/(A-cm).



However, only one polarity will generate a MOT, and that polarity depends on the laser polarization, which is determined by the user and the specific laser used. When this polarization is known, one can calculate which direction current should flow through the coils to produce a MOT.

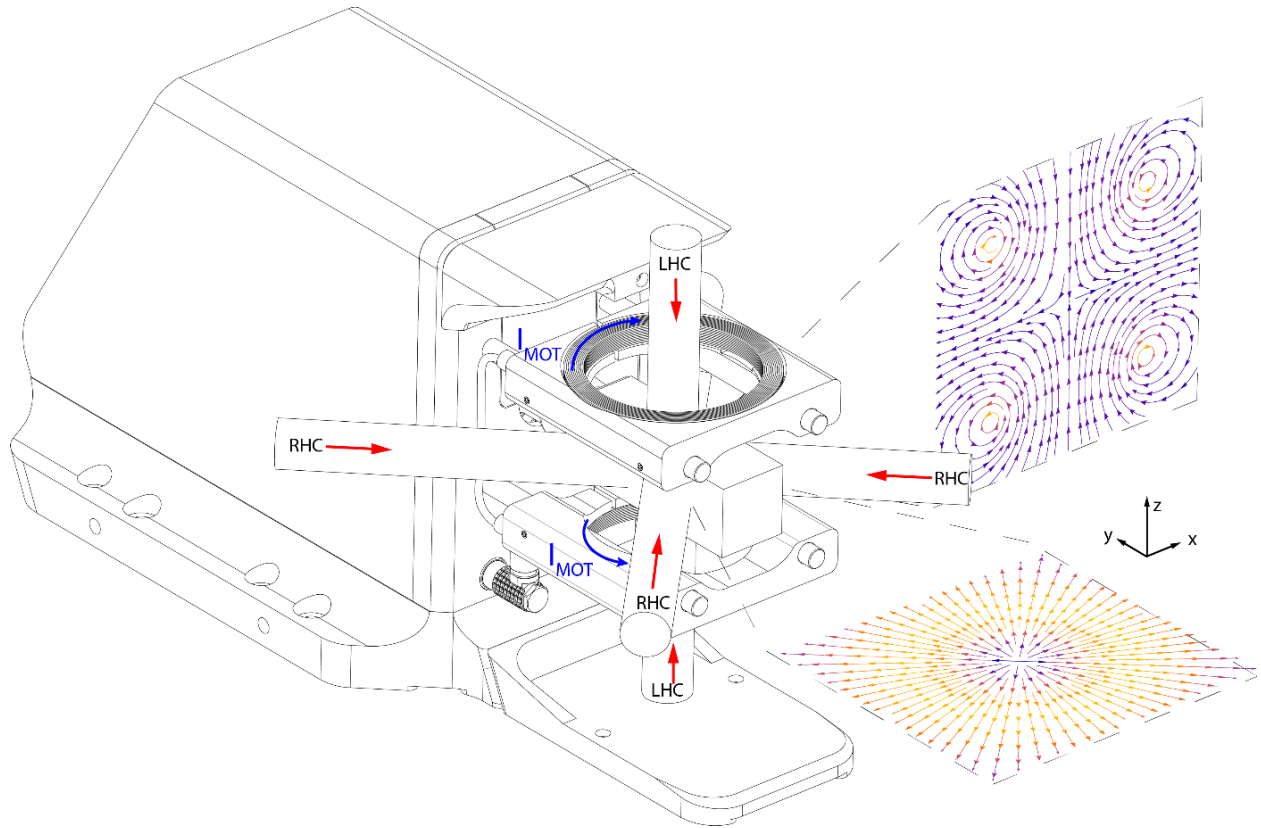


Figure 3. Three-dimensional plot of the magnetic field for miniMOT, shown with positive polarity through the coils. Correct optical polarization states are shown for the MOT beams, where RHC and LHC indicate right-hand circular and left-hand circular polarization respectively. In the case of negative polarity through the coils the magnetic field vector directions would reverse, and the handedness of the optical fields would also flip.



MA-00039

miniMOT V2 USER MANUAL

REVISION CONTROL

When any part of this procedure requires amendment, the document shall be re-issued in its entirety; requests for change shall be addressed to the document owner.

Revision	Date	Change Description	ECO	Originator(s)
Previous	May 2023	• Updated for miniMOT V2 launch • New Inflection branding		E. Salim
1.0	September 2025	• New format and new Inflection branding • Technical drawings updated	S1025	C. Ward C. Williams